

PAVAN DHARMA ADAPA

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PROFESSIONAL SUMMARY

Healthcare Data Analyst with an M.S. in Computer Science & Engineering, **available immediately**. Designs production ETL pipelines, dimensional data warehouses, and clinical-operations reporting across SQL, Python, Tableau, and Power BI. Portfolio work runs on **real public health data** — CDC NHAMCS emergency department microdata, CMS claims-denial filings, and multi-hospital inpatient encounters — with design-correct survey inference, calibrated risk models, and terminology-driven quality measures. Grounded in EHR/EMR concepts, HL7, ICD-10/CPT, SNOMED CT, LOINC, and HIPAA, and experienced translating clinical questions into validated data models and provider-facing reports mirroring Epic Clarity, Cogito, and Caboodle workflows.

TECHNICAL SKILLS

Epic Ecosystem & Reporting (self-study / certification prep): Clarity data model (ENCOUNTER, PAT_ENC, ORDER_PROC, RESULT, PAT_DIAGNOSIS), Cogito, Caboodle, Reporting Workbench concepts, SlicerDicer, Radar dashboards; Crystal Reports, SSRS, SAP BusinessObjects; Epic Sphinx assessment prep.

Healthcare Data & Standards: EHR/EMR systems, clinical & claims data, revenue cycle (denials, appeals, A/R, charge capture), population health, patient registries, quality measures (HEDIS, CMS eCQM), HL7 v2, ICD-10/CPT, SNOMED CT, LOINC, HIPAA compliance.

Data Engineering & Governance: ETL/ELT pipeline development, data warehousing, dimensional / star-schema modeling, grain design, data validation, cleansing & standardization, data quality, data governance.

Programming & Querying: SQL (joins, CTEs, window functions, query optimization), Python (Pandas, NumPy, scikit-learn, SciPy), PostgreSQL, MySQL, SQLite, Advanced Excel.

BI & Visualization: Tableau, Microsoft Power BI, Matplotlib, Seaborn — interactive dashboards, KPI reporting, executive-ready visualization.

Analytics & Modeling: Complex-survey estimation (weighting, ultimate-cluster variance, Taylor linearisation), hypothesis testing, EDA, risk stratification, predictive modeling (logistic regression, Random Forest, XGBoost, gradient boosting), model calibration, decision curve analysis.

Tools & Cloud: Git, Linux, AWS, Jupyter, Selenium, SQLite.

PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant & Analytics Mentor

Sep 2025 – May 2026

Mississippi State University — Starkville, MS

- Trained and supported **100+ students** across 6 courses on data cleaning, transformation, visualization, and dashboard-based reporting in SQL, Python (Pandas, Matplotlib), and Advanced Excel.
- Enabled early identification of at-risk students by tracking KPIs (completion rates, performance distribution, retention) in SQL and Excel, improving intervention turnaround.
- Built and led hands-on workshops on IoT sensor data for the SmartCT smart-city initiative, producing traffic, energy, and environmental KPI dashboards while reinforcing data-quality and validation practices.
- Created reusable training materials and documentation that improved task-completion speed by **25%** and standardized analytics best practices across cohorts.

Graduate Research Analyst — Healthcare Data Analytics

Aug 2024 – Aug 2025

Mississippi State University — Starkville, MS

- Designed and automated a production ETL pipeline (Python, Selenium, SQL) to extract, cleanse, and standardize multi-source clinical and surveillance datasets, cutting data-processing time **95%** — from several hours to under 10 minutes per dataset.
- Partnered with cross-functional stakeholders to translate clinical and operational questions into technical requirements, querying **900,000+** public-health surveillance records in SQL and Python to surface **8 key drivers** of disease transmission.
- Built reproducible, enterprise-scale reporting by engineering dimensional / star-schema data models and enforcing field-level validation, standardization, and governance across multiple source systems.
- Developed risk-stratification and predictive models (logistic regression, Random Forest, XGBoost) reaching **87% forecast accuracy**, and established baseline performance KPIs to monitor model and data quality over time.
- Reduced manual analysis time **50%** and resolved **35%** of dataset inconsistencies by automating feature-engineering and data-cleansing routines.
- Delivered presentation-ready Tableau and Power BI dashboards and correlation analyses that translated complex findings into actionable insight for leadership.

HEALTHCARE ANALYTICS PROJECTS

Emergency Department Throughput Analytics · CDC/NCHS NHAMCS — real national survey microdata ·

github.com/adapapavandharma/ed-throughput-analytics

- Analyzed **16,025 real emergency department visits**, survey-weighted to the **155.4M** US ED visits of 2022; the pipeline self-validates by reproducing CDC's published national total exactly or refusing to run.

- Implemented the **ultimate-cluster variance estimator with Taylor linearisation** CDC prescribes for this stratified multi-stage sample; design-correct inference showed the hypothesised peak-hour effect does **not** clear significance at $\alpha = 0.01$ — a negative result reported rather than discarded.
- Found between-hospital variation in median door-to-provider time (2–74 min) to be **3.9× larger** than time-of-day variation, and quantified boarding at **48.6M ED bay-hours lost annually** as the recoverable capacity item.
- Derived the 2,382-byte record layout by parsing CDC's 268-page documentation PDF rather than hardcoding byte offsets, with structural validation so a schema change fails loudly instead of silently mis-reading fields.

30-Day Readmission Risk & Care-Management Registry · 99,343 real inpatient encounters, 130 hospitals ·

github.com/adapapavandharma/readmission-risk

- Built a gradient-boosted readmission model (**AUC 0.664**) on real multi-hospital encounter data; top-quintile targeting captures **37.7% of all readmissions** at **1.89× enrichment**, number-needed-to-screen 4.8, delivered as a capacity-derived tiered risk board.
- Benchmarked against a single-variable clinical rule to isolate marginal value — prior inpatient count alone reaches 32.2% recall, so the model buys **+5.5 points**, materially less than an AUC comparison implies.
- Showed balanced class weighting inflates mean predicted risk to **46% against an observed 11%** while leaving AUC unchanged, establishing calibration rather than discrimination as the constraint on deployability.
- Handled the dataset's three traps explicitly: removed deceased and hospice discharges who cannot be readmitted, kept >30-day returns as hard negatives, and grouped every split on patient to prevent leakage.

Claims Denials & Appeals Analytics · CMS Transparency in Coverage — 3-year national panel ·

github.com/adapapavandharma/denials-analytics

- Built a 15,545 plan-year panel covering **126.9 million denied claims** across every Qualified Health Plan on the federal Marketplace, 2021–2023.
- Established that only **0.168% of denials are ever appealed** while **40.1% of appeals succeed**, identifying capacity to file — not claim merit — as the binding constraint on denial recovery.
- Showed the payer-reported reason taxonomy cannot support root-cause work ("**Other**" is **71.7%** of categorised denials and growing), and that denial rate is an organisational property: **6.0% to 48.3%** across issuers with 100,000+ claims.
- Modelled two fact grains because CMS repeats issuer-level fields across every plan an issuer sells, and classified suppression markers before coercion — a value withheld for a small cell is missing-not-at-random, so averaging the survivors biases every aggregate upward.

Clinical Quality Measures Engine · HEDIS / CMS eCQM over a 13,810-patient EHR · github.com/adapapavandharma/clinical-quality-measures

- Built a declarative measure engine evaluating HEDIS and CMS eCQM specifications over **820,993 encounters and 10.8M observations**; measures are versioned YAML specs with SNOMED and LOINC value sets, so an annual steward update is a reviewable diff rather than a code change.
- Decomposed CMS122 to show its 17.3% poor-control score comprised **1 patient failing on a recorded result and 203 never tested**, reclassifying the finding from a glycaemic-control problem to a testing-outreach problem.
- Exposed a **32-point gap** in blood pressure control between ages 18–44 (57%) and 75+ (89%) hidden by the 66.9% aggregate, and delivered a care-gap registry of 4,439 gaps prioritising the **915 patients with two or more** closable in one contact.

Multimodal LLM Evaluation (LLaVA-1.5) · Analytics & experimentation — academic research

- Designed evaluation pipelines analysing cross-lingual performance of multimodal large language models, applying NLP techniques, data-contamination auditing, and statistical analysis across multiple benchmarks.
- Aggregated experiment metrics in Python and SQL into clean analytical datasets enabling systematic comparison across models, hyperparameters, and prompts, and presented findings to academic stakeholders.

EDUCATION

M.S., Computer Science & Engineering

Graduated May 2026

Mississippi State University — Mississippi State, MS

Coursework: Advanced Machine Learning, Artificial Intelligence, Algorithms, Parallel Algorithms, Software Engineering, Data Analytics, Advanced Cyber Operations.

B.Tech, Computer Science & Engineering

Graduated May 2024

Malla Reddy University — Hyderabad, India

Coursework: Data Structures & Algorithms, DBMS, Statistics & Probability, Machine Learning, Operating Systems, Computer Networks, OOP.

PUBLICATION (IN PROGRESS)

Computational Tools for Modeling and Predicting Respiratory Disease Spread in Commercial Poultry Production

Zhang Li, Pavan Dharma Adapa, Sai Harika Gade, Saikanth Ratnavale, Navicky Michael.

- Designed forecasting frameworks synthesizing compartmental (SIR/SEIR) and machine-learning models to predict respiratory disease transmission at scale, and assessed Random Forest and deep-learning methods for identifying risk factors and detecting anomalies in surveillance data.